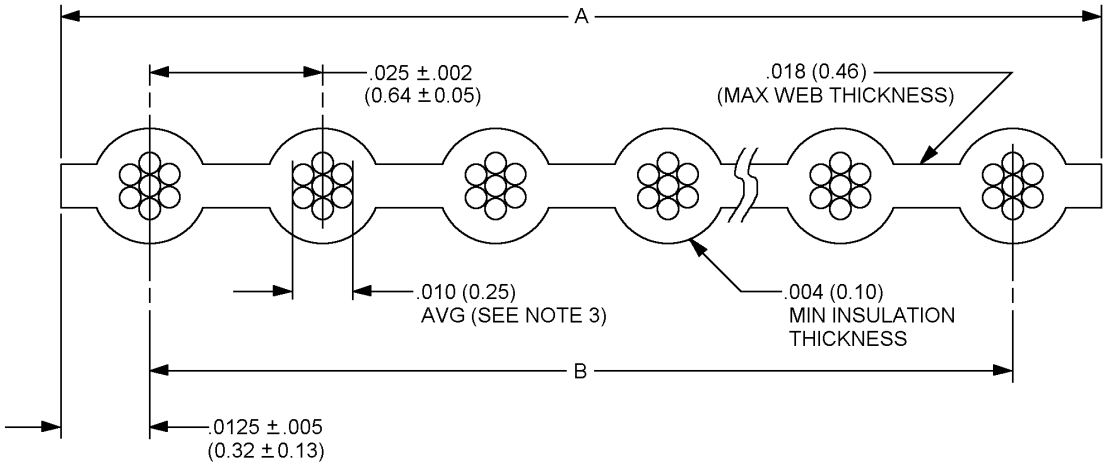
INCH-POUND
MIL-DTL-49055/21
w/ AMENDMENT 1
9 June 2017
SUPERSEDING
MIL-DTL-49055/21
14 September 2004

DETAIL SPECIFICATION SHEET

CABLES, SPECIAL PURPOSE, FLEXIBLE, FLAT, UNSHIELDED, 32 AWG STRANDED, ROUND CONDUCTOR 200°C (.025 INCH PITCH)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49055.



Inches	mm
.002	0.05
.004	0.10
.005	0.13
.010	0.25
.0125	0.318
.025	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Theoretical average diameter for stranded wires.

FIGURE 1. Dimensions and configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics.

Dash number	Number of Conductors	Cable width 'A' inches	End condition to end condition "B" inches	Weight (lb/1,000 ft)	
				Nominal	Max
01	10	.250 ± .007	.225 ± .005	7.0	10%
02	16	.400 ± .007	.375 ± .005	11.2	10%
03	20	.500 ± .007	.475 ± .005	14	10%
04	26	.650 ± .007	.635 ± .005	18.2	10%
05	34	.850 ± .007	.825 ± .005	23.8	10%
06	40	1.000 ± .007	.975 ± .006	28	10%
07	50	1.250 ± .010	1.225 ± .006	35	10%
08	60	1.500 ± .010	1.475 ± .006	42	10%
09	64	1.600 ± .010	1.575 ± .006	44.8	10%
10	68	1.700 ± .010	1.675 ± .006	47.6	10%
11	80	2.000 ± .010	1.975 ± .006	56	10%
12	100	2.500 ± .010	2.475 ± .006	70	10%

REQUIREMENTS:

Dimensions and configurations: See figure 1 and table I.

Insulation material: Extruded Fluorinated Ethylene Propylene (FEP).

Temperature rating: -55°C to +200°C.

Voltage rating: 300 V rms at sea level.

DC resistance: 166 ohms/1000 feet (304.8 m) at 20°C.

Conductor:

AWG: 32 AWG.

Stranding: 7/40.

Conductor coating: Silver plated copper.

Withstanding voltage (at sea level): 2,000 V rms.

Thermal shock: -55°C to +200°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

Flammability: Non-burning.

Flexing endurance (-55 ± 5°C to + 200 ± 5°C): 500 cycles.

Moisture resistance: When the moisture resistance test is performed as specified in MIL-DTL-49055 the insulation resistance shall not be less than 50 megohoms per 1,000 (304.8 m) feet minimum.

Shrinkage: .125 inch (3.18 mm) on either end.

Markings (optional): In accordance with MIL-DTL-49055.

Strippability: Not applicable.

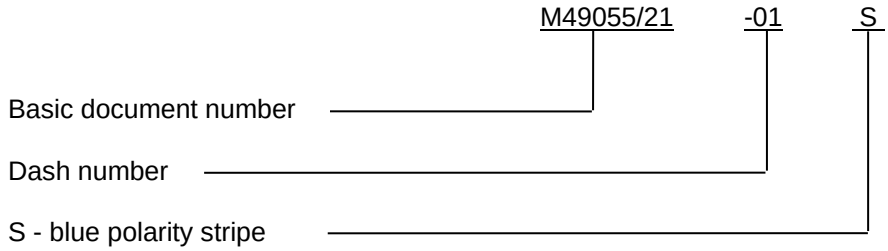
Solderability: Not applicable.

Tear groove propagation: Not applicable.

Color: Natural with red polarity stripe

Part or Identifying Number (PIN): The PIN shall consist of the letter M, the basic number of this specification sheet, the dash number from table I and “S” for polarity stripe.

Example:



Qualification is not required. When specified in the contract, a sample shall be subjected to first article inspection. First article inspection shall consist of group A and group B testing.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced document. This document references MIL-DTL-49055.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2017-017)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.